

DETECTIVE DATA

THE LONG ARM OF THE LAW JUST GOT A LITTLE LONGER THANKS TO BIG DATA.
BUT ARE YOU WILLING TO LET GO OF YOUR PRIVACY TO BE SAFER?

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Imagine walking down the street in the near future, the IoT has reached its peak and its ever pervasive nature entails that anything and everything in and around you is connected, and petabytes of data is being sent from trillions of connected systems to intensely powerful analytical systems. You are decked with gadgets – smartphone, smartband, and an AR visor are bare minimum.

As you continue walking, your AR (augmented reality) visor has your morning schedule listed and its navigation assistant directs you to take a turn to save a few minutes. You turn into the corner and your visor camera analyses every object and every person in front of you, highlighting those whom you may be connected to on social media. Your visor immediately turns a shade of orange and outlines a man 150 meters ahead of you. Your visor camera hasn't identified the person but the mere fact that he has been pacing to and fro rather hurriedly gets flagged. You notice that some of the CCTV cameras on the street turn in his direction.

Seconds later, the orange alert turns red and so does his outline. You become alert and come to a halt, the visor navigation directions reroute to show you the fastest way to safety. You notice that every other person on the street just received the same alert on their visors and they've all started moving away from the suspicious person. A counter soon pops up on your visor, the cops are on their way and they'll reach in barely two minutes. The area you're in has a heightened police density owing to the number of crime incidents here. You've double-backed to the earlier street corner, notice that the cops are already there and they're already

observing the person. A crime has not taken place so they can't arrest anyone. Two officers from the other end of the street slowly walk towards the suspect with a smile on their face...

It might seem a little too futuristic, but similar systems have already been designed and multiple pilot programmes are being run in cities and towns across the globe. Predictive Crime Analytics is very much in the present thanks to Big Data.

PREDICTIVE CRIME ANALYTICS

While not quite the way *Minority Report* – a Tom Cruise movie about predictive crime – pictured it, quite a few software already deal with Predictive Crime Analytics with even technology giants such as IBM and Hitachi coming out with their systems. Needless to say, quite a lot of questions have been raised regarding the efficacy of such systems. PCA is doing exactly what analysts and detectives have been doing for decades – poring over all the evidence to identify patterns and shortlist suspects. The only difference is that our methods of obtaining information have multiplied to the extent that humans will soon not be able to handle the enormous amount of data that needs to be analysed.

ENTER, MACHINE LEARNING.

Traditionally, databases would be populated manually by humans so there was a defined structure but this would also mean that the very same structure ended up being restrictions to the kind of data that could be stored in the database and thus a portion of the data would be discarded. As technology progressed,